AMIT RANJAN SAHU

Product Development Engineer, Sandisk
M.Tech, Indian Institute of Technology Bhubaneswar

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INDUSTRY EXPERIENCE

SANDISK- Product Development Engineer *(July'26-Current)*

ACADEMIC QUALIFICATION

Course	College/University	Major	Year	CGPA
M.Tech	Indian Institute Of Technology Bhubaneswar	Semiconductor Technology & Chip Design	2024–26	8.15/10
B.Tech	Dr. A.P.J. Abdul Kalam Technical University	Electronics & Communication Engineering	2018–22	7.08/10

KEY COURSES AND TECHNICAL SKILLS

EDA Tools & Software: Sentaurus TCAD LTspice Cadence Virtuoso.

Language: Python(NmuPy) C++ Verilog

Hardware & **Instruments**: Static thermo-electrical characterization Keithley power SMUs XRD FESEM AFM Gas Flow & Sensing Characterization

Course Work:

- Analog CMOS VLSI Design
- Digital CMOS VLSI Design
- Semiconductor Device Fabrication
- Power Semiconductor Devices
- Architecture Design of Digital ICs
- Semiconductor Device Packaging

PUBLICATIONS

- **Patent:** A novel edge termination technique for CSSJ diode with improved area efficiency and performance**(In process of filing)**
- **Publication**: TCAD Based Study of 4H-SiC Charge Sheet Superjunctions with a Wide Trench Based Termination, PCIM India 2026 **(Accepted for oral presentation)**

M.TECH PROJECTS

TCAD BASED STUDY OF 4H-SiC CSSJ WITH EDGE TERMINATION *(Sept'25–Apr'26)*

- 2D Design and Simulation of 4H-SiC CSSJ diode for a target $\mathbf{V}_{BR}$ **of 1.7*KV*** using Hatakeyama model in Sentaurus TCAD.
- Introduction of **novel charge sheet trench termination** & comparison with conventional termination.

STEADY STATE ELECTRO-THERMAL CHARACTERIZATION OF POWER MOSFET | iVP SEMICONDUCTOR *(Jan'26–Apr'26)*

- **Instruments**: **Keithley power SMUs(26XX series)**, DC power supply, High temperature chamber.
- Extraction of critical Static device parameters like threshold voltage,breakdown voltage, drain current of MOSFET(iVP semiconductor) at different temperature values.
- Python code for gate charge extraction for MOSFETs

PARKING LOT OCCUPANCY COUNTER |M.Tech-Term Project *(Jan'25–Apr'25)*

- Design of FSM with two input signals comes from Photo sensors, Design of HDL Code.

ANALOG AND DIGITAL CIRCUIT DESIGN| M.Tech-Term Project *(Jul'24–Jan'25)*

- Circuit, Layout design of multiple analog and digital circuits.

POSITIONS OF RESPONSIBILITY

Instructor for Semiconductor and Chip Design Workshop *(Dec'25)*

- Served as instructor for Semiconductor and Chip Design Workshop conducted organized by Edu Fabrica at IIT Bhubaneswar

Teaching Assistant for VLSI Design Lab, IIT Bhubaneswar *(Jan'26–Mar'26)*

- Assisted class of B.Tech 3rd Year in circuit simulation,layout design in cadence virtuoso software.

CERTIFICATIONS

- **ISWDP Cohort-4 by IISc in collaboration with SYNOPSYS and SAMSUNG(SSIR)** *(Apr 2025 -Jun 2025)*: **Received Grade-I Scholarship for Level-1, Level-2 and Advance** Structure and Process simulation of semiconductor devices(Diode, MOSFET) using Sentaurus TCAD for the real world application